

YOUR
LOGO

SmartHire AI

2026-07-21 |
VERSION 1.0 |
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Project Introduction

SmartHire AI helps job seekers improve resumes. Upload PDF/DOCX for AI analysis.

CORE CAPABILITIES

- Resume parsing
- ATS Analysis
- Resume Scoring
- Skills gaps
- Strengths/weaknesses
- AI Suggestions
- Interview Questions
- Readiness Score
- MySQL storage
- Recruiter dashboard

TECHNOLOGY

React.js, Node.js, Express, MySQL, Google Gemini AI, AWS

HIGHLIGHTS

- PDF/DOCX
- AI analysis
- Suggestions

KEY BENEFITS

- Professional feedback
- Affordable
- Real-time ATS scoring
- Role-specific tips

ARCHITECTURE

- React.js frontend
- Node.js backend
- MySQL database
- Gemini AI
- AWS Cloud

Executive Summary

Platform

End-to-end AI recruitment intelligence solution bridging the gap between job seekers and employment success

Market

Primary: Students, Fresh Graduates, Job Seekers, Career Switchers
Secondary: Recruiters, HR Professionals

Value

Democratizing access to professional resume optimization through AI
Freemium SaaS model

The Modern Job Search Crisis

No affordable solution exists that provides comprehensive, AI-driven resume intelligence for candidates struggling in today's competitive job market

01

75% of resumes rejected by ATS before human review. Job seekers lack visibility into failures.

02

Professional resume services cost \$200-500. Fresh grads face experience catch-22.

03

Recruiters spend 23 hours per hire screening unqualified candidates.

Existing Recruitment Challenges

FOR JOB SEEKERS

- Black box application process with zero feedback
- Generic resume templates that fail ATS parsing
- No understanding of keyword optimization
- Expensive career coaching (\$150-300/hour)
- Interview anxiety from lack of preparation

MARKET GAP

No unified platform addresses both candidate preparation AND recruiter efficiency. Job seekers lack affordable tools while recruiters waste time on manual screening.

FOR RECRUITERS

- Manual resume screening consumes 60% of hiring time
- Inconsistent evaluation criteria across team members
- High volume of unqualified applications
- Difficulty identifying hidden talent among applicants

Why SmartHire AI

Unique Value Propositions

- 1

Candidate-First Philosophy
Built specifically to empower job seekers, not just serve recruiters
- 2

AI-Powered Intelligence
Google Gemini integration delivers human-quality analysis at scale
- 3

Comprehensive Analysis
Single upload generates ATS score, improvement suggestions, and interview prep
- 4

Actionable Insights
Every recommendation is specific, implementable, and role-relevant

- 5

Dual-Sided Platform
Candidates improve; recruiters receive pre-analyzed, higher-quality applicants
- 6

Cost Accessibility
Professional-grade analysis available without expensive consultant fees
- 7

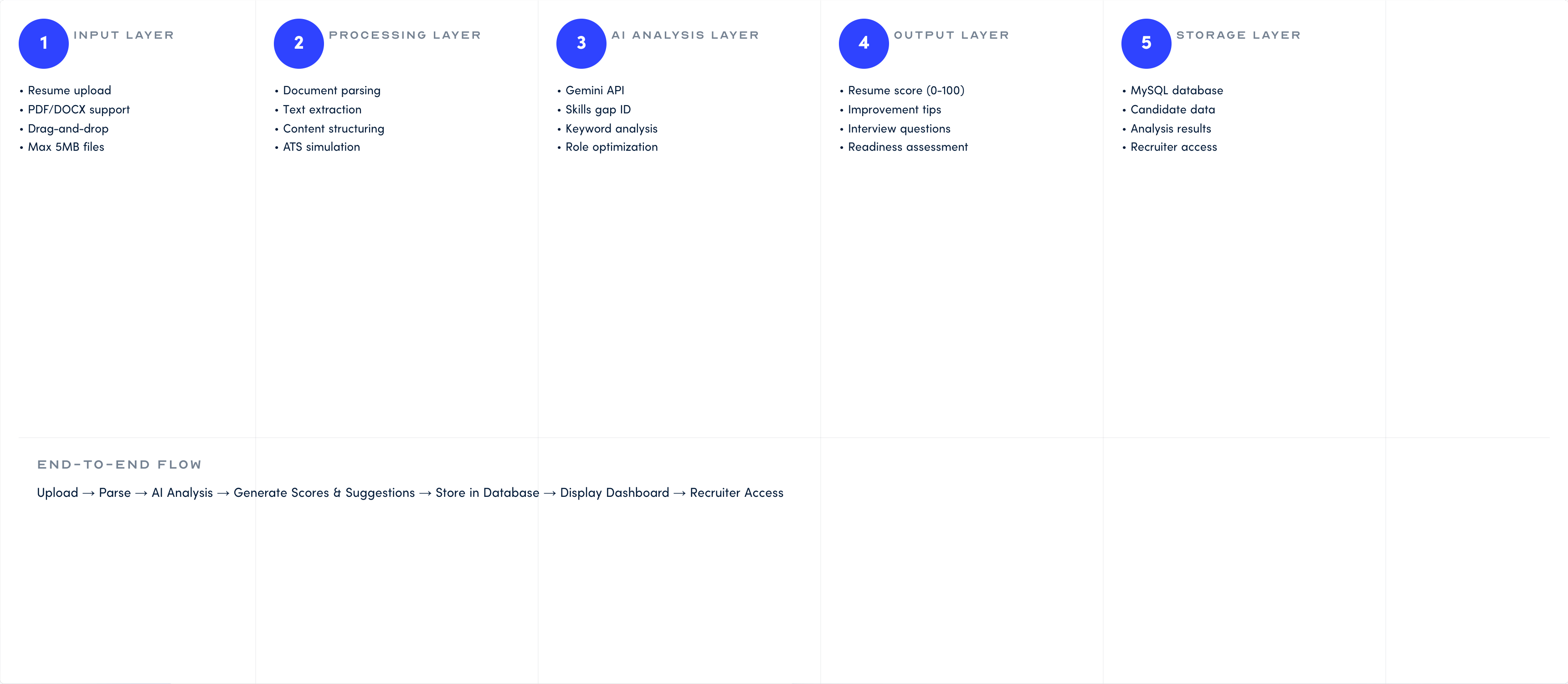
Portfolio-Quality Architecture
Enterprise-grade technology stack demonstrating production readiness

THE SMARTHIRE ADVANTAGE

SmartHire AI uniquely positions itself at the intersection of candidate empowerment and recruiter efficiency. By providing AI-powered analysis at an accessible price point, we democratize professional resume optimization while creating a pipeline of better-prepared candidates for hiring teams.

Proposed Solution

SmartHire AI Solution Framework



Product Features - Candidate Experience

UPLOAD & PARSING

- Landing page with clear value proposition and CTA
- Drag-and-drop resume upload (PDF/DOCX, max 5MB)
- Intelligent extraction of contact info, experience, education, skills

SCORING & INSIGHTS

- ATS Compatibility Score (0-100)
- Detailed breakdown of pass/fail criteria
- Strengths and weaknesses highlighting
- Keyword gap analysis

INTERVIEW PREPARATION

- 10-15 custom questions based on resume content
- Questions tailored to target role
- Comprehensive preparedness assessment
- Interview Readiness Score

ANALYSIS FEATURES

- AI-generated executive summary of candidate profile
- Industry and role-specific keyword identification
- Technical and soft skills mapping with proficiency indicators

RECOMMENDATIONS

- Prioritized, actionable improvement suggestions
- Section-by-section guidance
- Role-specific optimization tips

EXPORT & ACTIONS

- Download complete analysis as PDF report
- Offline reference for interview prep
- Re-upload improved resume for iteration

COMPLETE CANDIDATE WORKFLOW

From resume upload to interview readiness in minutes: Upload → Parse → Analyze → Score → Suggest → Prepare → Export. SmartHire AI provides an end-to-end experience that transforms job seekers into confident, well-prepared candidates.

Product Features – Recruiter Dashboard

RECRUITER PORTAL FEATURES



Candidate Dashboard: Centralized view of all analyzed candidates in one place



Candidate Reports: One-click access to complete candidate analysis and insights



Resume Summary: Quick-read AI-generated candidate overview for fast screening



ATS Score Display: Instant visibility into candidate ATS compatibility metrics



Interview Score Access: Pre-computed interview readiness metrics and generated questions

VALUE FOR RECRUITERS

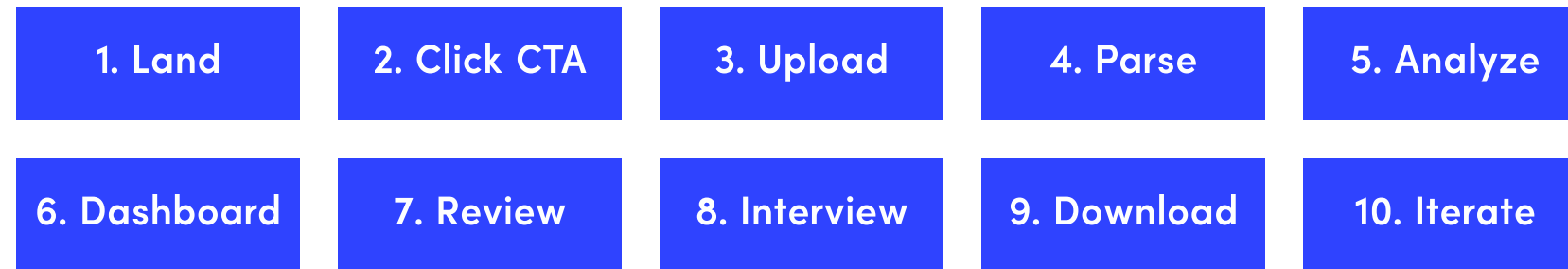
- Receive pre-screened, self-improved candidates
- Reduce initial screening time by 40%
- Standardized scoring enables objective comparison
- Interview questions pre-generated for each candidate
- Access comprehensive candidate analytics

KEY BENEFIT

Recruiters benefit from a pipeline of candidates who have already optimized their resumes and prepared for interviews, dramatically reducing time-to-hire and improving candidate quality.

User Journey

CANDIDATE JOURNEY



RECRUITER JOURNEY



JOURNEY DETAILS

Candidate Flow:

1. Land on SmartHire AI homepage
2. Click "Analyze My Resume" CTA
3. Upload resume (PDF/DOCX)
4. System parses and extracts content
5. AI analyzes resume against ATS criteria
6. View comprehensive dashboard with scores
7. Review improvement suggestions
8. Access custom interview questions
9. Download complete report as PDF
10. Iterate and re-upload improved resume

Recruiter Flow:

1. Access recruiter dashboard
2. Browse analyzed candidates
3. View individual candidate reports
4. Use pre-generated interview questions

Recruiters benefit from pre-screened candidates who have already optimized their profiles.

System Architecture

Three-Tier Architecture

Presentation Tier (Frontend)

- React.js SPA with component-based architecture
- Tailwind CSS for responsive, utility-first styling
- Axios for HTTP client communication
- React Router for client-side navigation

Application Tier (Backend)

- Node.js runtime environment
- Express.js RESTful API framework
- Multer middleware for file upload handling
- Google Gemini API integration for AI processing

Data Tier (Database)

- MySQL relational database
- Structured storage for candidates, analyses, scores
- AWS RDS managed database service
- Automated backups and scaling

COMMUNICATION FLOW

Client → REST API → Business Logic → AI Service → Database → Response



Technology Stack

Frontend Technologies

- React.js 18: Component-based UI library
- Tailwind CSS: Utility-first CSS framework
- Axios: Promise-based HTTP client
- React Router v6: Declarative routing

Backend Technologies

- Node.js: JavaScript runtime
- Express.js: Minimal web framework
- REST API: Stateless API architecture
- Multer: Multipart form data handling

Cloud Infrastructure

- AWS EC2: Compute instances
- AWS RDS: Managed MySQL database
- VPC: Secure network configuration
- Elastic IP: Static addressing

Database

- MySQL 8.0: Relational database management
- Optimized queries for performance
- Structured schema design

Artificial Intelligence

- Google Gemini API: Large language model
- Intelligent resume analysis
- Natural language processing

Development Tools

- Git: Version control
- PM2: Process manager
- ESLint: Code quality

FUTURE ENHANCEMENTS

Amazon S3 for file storage | Docker containerization | CloudFront CDN | CI/CD Pipeline with GitHub Actions

Database Design Overview

Core Database Tables

1

CANDIDATES

candidateid (PK), name, email, phone, uploadedat, resumepath

2

RESUMES

resumeid (PK), candidateid (FK), originalfilename, parsedcontent, uploadtimestamp

3

ANALYSIS_RESULTS

analysisid (PK), resumeid (FK), atsscore, keywordscore, skillsextracted, strengths, weaknesses, createdat

4

IMPROVEMENT_SUGGESTIONS

suggestionid (PK), analysisid (FK), section, suggestiontext, prioritylevel

5

INTERVIEW_QUESTIONS

questionid (PK), analysisid (FK), questiontext, category, difficultylevel

6

INTERVIEW_SCORES

scoreid (PK), analysisid (FK), readinessscore, breakdownjson

DATABASE RELATIONSHIPS

One-to-Many: candidates → resumes | One-to-One: resumes → analysisresults | One-to-Many: analysisresults → improvementsuggestions, interviewquestions, interviewscores

ENTITY RELATIONSHIP (ERD)

The database follows normalized design principles with clear foreign key relationships. Candidates can have multiple resumes, each resume has one analysis result, and each analysis generates multiple suggestions, questions, and scores. This structure enables efficient querying for both candidate dashboard views and recruiter reports.

AI Workflow

Google Gemini AI Integration Pipeline

1

INPUT PROCESSING

- Resume text extraction from PDF/DOCX
- Content cleaning and normalization
- Section identification (Experience, Education, Skills)

2

PROMPT ENGINEERING

- Role-specific analysis prompts
- ATS simulation criteria injection
- Structured output formatting

3

AI ANALYSIS

- ATS compatibility assessment
- Keyword density calculation
- Skills gap identification
- Strength/weakness classification

4

RESPONSE PROCESSING

- JSON parsing of AI responses
- Score normalization (0-100)
- Suggestion prioritization
- Interview question generation

5

STORAGE

- Persist all results to MySQL
- Enable retrieval for dashboards
- Recruiter access provisioning

AI PROCESSING HIGHLIGHTS

- Google Gemini provides state-of-the-art natural language understanding
- Prompts engineered for consistent, structured JSON output
- Real-time processing with average response time under 5 seconds
- Contextual analysis based on industry and role specifications
- All AI responses validated before storage
- Error handling ensures graceful degradation
- Scoring algorithms calibrated against industry benchmarks
- Continuous improvement through feedback loops

AWS Deployment Architecture

Production Infrastructure

Compute Layer

- AWS EC2 t2.micro
- Ubuntu 22.04
- Node.js server
- PM2 manager

Database Layer

- AWS RDS MySQL
- db.t3.micro
- Auto backups
- Multi-AZ

Network

- VPC

Security

- HTTPS
- Env vars
- IAM

Future

- S3
- CDN
- Docker
- CI/CD

Security and Performance

Security

- Input validation
- File verification
- SQL prevention

Performance

- Async patterns
- Connection pooling
- Compression
- Lazy loading

Monitoring

- PM2 monitoring
- CloudWatch
- Error logging
- Alerts

SECURITY-FIRST

Defense-in-depth security. All data encrypted in transit and at rest.

Expected Outcomes and Business Value

Technical Outcomes

- Fully functional AI recruitment platform
- Production-deployed AWS application
- RESTful API with 10+ endpoints
- Responsive React.js frontend
- AI-integrated analysis pipeline

Business Value

- Reduces candidate resume optimization time by 80%
- Decreases recruiter screening time by 40%
- Democratizes access to professional resume feedback
- Measurable improvement in application success rates

Portfolio Value

- Demonstrates full-stack development proficiency
- Showcases AI/ML integration capabilities
- Proves cloud deployment expertise
- Exhibits software engineering best practices

KEY IMPACT METRICS

For Candidates:

- 80% reduction in resume optimization time
- Professional-grade feedback at accessible cost
- Improved interview confidence

For Recruiters:

- 40% reduction in screening time
- Higher quality candidate pipeline
- Standardized evaluation metrics

For Portfolio:

- End-to-end project completion
- Production-ready deployment
- Modern tech stack expertise

Future Roadmap and Conclusion

VERSION 2.0

- User authentication (JWT-based)
- Job posting and matching
- Advanced analytics dashboard
- Resume version history
- Amazon S3 file storage migration
- Docker containerization
- CI/CD pipeline automation

VERSION 3.0

- Applicant tracking system integration
- Payment gateway for premium features
- Admin dashboard for platform management
- Email notifications
- Advanced search and filtering

CONCLUSION

SmartHire AI represents the intersection of AI innovation and practical job seeker needs. Built with enterprise-grade technology stack and deployed on production-ready AWS infrastructure.

PROJECT SUMMARY

SmartHire AI delivers a comprehensive AI-powered recruitment intelligence platform that empowers job seekers while improving recruiter efficiency. The platform demonstrates:

- Full-stack development proficiency (React, Node.js, MySQL)
- AI/ML integration expertise (Google Gemini)
- Cloud deployment capabilities (AWS EC2, RDS)
- Software engineering best practices
- Production-ready architecture and deployment

CONTACT

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Version 1.0

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